

EQUIDISTRIBUTION OF COMMUTATORS IN FINITE SYMPLECTIC GROUPS

SOPHIE KRIZ

ABSTRACT. We prove the equidistribution of non-central commutators in symplectic groups over finite fields of odd characteristic.

1. INTRODUCTION

The commutator map is an object of significant interest in finite group theory. For example, one of the most celebrated results of the area is the proof by M. Liebeck, E. O'Brien, A. Shalev, and P. H. Tiep [8, 11] of the Ore conjecture, stating that for a finite non-abelian simple group, the commutator map is surjective (see [12]). However, in many cases, e.g. the finite reductive groups, the distribution of the commutators still remains mysterious, especially in the case of unipotent elements with large centralizers (see [6, 7, 14]). In this paper, we use the method of Frobenius sums to prove the equidistribution of non-central commutators in symplectic groups of a fixed rank over an increasing field, proving a conjecture of A. Shalev (see Conjecture 1.11 of [13]). This result is perhaps surprising, given the fact that an analogous equidistribution result for other groups is false in general (Shalev, Tiep [14], with a correction by Deligne [1]).

To be more specific, we recall the method of Frobenius sums: In this paper, we consider finite-dimensional complex representations ρ of finite groups G . For an element $g \in G$, we write $\chi_\rho(g)$ for the character value of ρ at g (i.e. the trace of g 's action on ρ). We write $\widehat{G}$ for the set of irreducible (non-zero) G -representations.

Then, for an element of a finite group $g \in G$, the number of pairs $(x, y) \in G \times G$ such that $xyx^{-1}y^{-1} = g$ can be calculated as the group order multiplied by the *Frobenius sum*:

$$(1) \quad |\{(x, y) \in G \times G \mid xyx^{-1}y^{-1} = g\}| = |G| \sum_{\rho \in \widehat{G}} \frac{\chi_\rho(g)}{\dim(\rho)}.$$

In particular, we see the trivial element is always produced in the maximal number of ways, with (1) outputting $|G| \cdot |\widehat{G}|$. We also note that the summand of (1) corresponding to the trivial representation $\rho = 1$

is always $|G|$, independent of g . In particular, the terms in (1) which actually depend on g sum to

$$(2) \quad \sum_{\rho \neq 1 \in \widehat{G}} \frac{\chi_\rho(g)}{\dim(\rho)},$$

which we call the *reduced Frobenius sum*. We can also then see that the commutators are fairly evenly distributed if (2) is very small for all elements excluding a small number of outlier cases like $g = I$, since this would give about $|G| = |G \times G|/|G|$ pairs per generic choice of g . Our main result can then be stated as the following

Theorem 1. *Fix an odd prime power q and fix a non-central element $g \neq \pm I$ in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$. The reduced Frobenius sum for g is then $o(1)$ with respect to q :*

$$(3) \quad \lim_{q \rightarrow \infty} \sum_{\rho \neq 1 \in \widehat{\mathrm{Sp}_{2N}(\mathbb{F}_q)}} \frac{\chi_\rho(g)}{\dim(\rho)} = 0$$

(we consider this limit and all others in this paper to be over a sequence of prime powers q^{n_i} such that $n_i | n_{i+1}$ and every natural number divides an eventual n_j).

Corollary 2. *The number of pairs $(x, y) \in \mathrm{Sp}_{2N}(\mathbb{F}_q) \times \mathrm{Sp}_{2N}(\mathbb{F}_q)$ such that $xyx^{-1}y^{-1} = g$, divided by the group order $|\mathrm{Sp}_{2N}(\mathbb{F}_q)|$, converges to 1 as q tends to ∞ as a natural number, uniformly in all odd prime powers q and all elements $g \in \mathrm{Sp}_{2N}(\mathbb{F}_q) \setminus Z(\mathrm{Sp}_{2N}(\mathbb{F}_q))$.*

Proof. (P. Deligne) Recalling (1), the vanishing (3) of the reduced Frobenius sum immediately implies the claimed convergence as n increases multiplicatively in the sense of Theorem 1.

Now let us consider the commutator map

$$c : \mathrm{Sp}_{2N} \times \mathrm{Sp}_{2N} \rightarrow \mathrm{Sp}_{2N}$$

as a map of schemes over $\mathbb{F}_q$. Denote the fiber at each point $g \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$ by X_g . The convergence obtained from Theorem 1 can instead be interpreted in terms of the number of points in X_g over each $\mathbb{F}_{q^n}$, stating that for $g \neq \pm I$, $|X_g(\mathbb{F}_{q^n})|/q^{n \dim(\mathrm{Sp}_{2N})}$ tends to 1 as n increases in the same multiplicative sense as in Theorem 1. This is then equivalent to $X_g \times_{\mathrm{Spec}(\mathbb{F}_q)} \mathrm{Spec}(\overline{\mathbb{F}_q})$ having exactly 1 irreducible component of dimension equal to $\dim(\mathrm{Sp}_{2N}) = 2N^2 + N$ (and no irreducible components of higher dimension) for all non-central $g \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$. Given this, the constructibility of $R^i c_! \underline{\mathbb{Q}}_\ell$ on Sp_{2N} over $\mathrm{Spec}(\mathbb{Z})$ implies

the statement using the Leray spectral sequence with compact supports and the Lefschetz Theorem. $\square$

The situation at $g = -I$ is described by the following

Proposition 3 (P. Deligne). *For q an odd prime power, the number of pairs of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ elements whose commutator is $-I$ is*

$$q^{\lfloor N/2 \rfloor} + O(q^{\lfloor N/2 \rfloor - 1}).$$

Proof. For any finite group G and $g \in Z(G)$, the number of pairs $(x, y) \in G^2$ where $xyx^{-1}y^{-1} = g$ is equal to $|G|$ times the number of conjugacy classes of elements $c \in G$ such that c is conjugate to cg . This follows since

$$xyx^{-1}y^{-1} = g$$

is equivalent to

$$x^{-1} = y^{-1}(x^{-1}g)y.$$

Therefore, for each conjugacy class (c) such that c and cg are conjugate, we get

$$|G| = |(c)| \cdot |Z_G(c)|$$

choices of (x, y) (where $Z_G(c)$ denotes the centralizer of c in G), by choosing x^{-1} to be any element of the conjugacy class (c) , in which case $|Z_G(c)|$ is exactly equal to the number of choices of y conjugating x^{-1} to $x^{-1}g$.

For $G = \mathrm{Sp}_{2N}(\mathbb{F}_q)$ and $g = -I$, to be conjugate to $cg = -c$, c must have a characteristic polynomial P which is monic and self-reciprocal, satisfying $P(z) = -P(z^{-1})$. There are $q^{\lfloor N/2 \rfloor}$ such polynomials. The fraction of such polynomials corresponding to semisimple elements without repeated eigenvalues is $1 - O(1/q)$, while the others contribute a lower power of q . This gives the claimed statement. $\square$

Our strategy of proof is to separate off the sum of character ratios corresponding to a certain special class $\mathcal{P}$ of *primitive representations*, rewriting the left hand side of (3) as the limit of two partial sums

$$(4) \quad \sum_{\rho \in \mathcal{P}} \frac{\chi_\rho(g)}{\dim(\rho)} + \sum_{\rho \in \widehat{\mathrm{Sp}_{2N}(\mathbb{F}_q)} \setminus \mathcal{P}} \frac{\chi_\rho(g)}{\dim(\rho)}.$$

A key property of $\mathcal{P}$ is that the number of irreducible representations is constant with respect to q (and therefore to prove the vanishing of the first sum in (4), it suffices to prove each individual character

ratio $\chi_\rho(g)/\dim(\rho)$ vanishes for $\rho \in \mathcal{P}$). On the other hand, the non-primitive representations can be reduced using Deligne-Lusztig induction.

To define the class of primitive irreducible representations, we recall that for a finite reductive group $G(\mathbb{F}_q)$, the irreducible representations can be classified according to the data of a semisimple conjugacy class (s) in the dual group $G^*(\mathbb{F}_q)$, a unipotent irreducible representation u of (the dual of) the identity component of s 's centralizer, and possible additional data in cases when G or $Z(G)$ is disconnected. The primitive irreducible representations are then defined as follows:

Definition 4. *Consider a finite reductive group $G(\mathbb{F}_q)$. We say an irreducible $G(\mathbb{F}_q)$ -representation ρ is primitive if the semisimple conjugacy class (s) in the dual group $G^*(\mathbb{F}_q)$ which is a part of the classification data of ρ has only ± 1 eigenvalues (i.e. $s^2 = I$). We write $\mathcal{P}_G$ for the set of irreducible primitive G -representations (we omit the subscript when G is fixed).*

Equivalently, primitive irreducible representations are precisely those appearing with non-zero coefficient in the Deligne-Lusztig inductions of torus characters with trivial squares. For example, every unipotent irreducible representation is primitive, and in the case of G a symplectic group, the oscillator representation pieces are also primitive. We also note that, like the unipotent representations, the number of primitive irreducible representations, say of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ or $\mathrm{O}_n(\mathbb{F}_q, \pm)$, does not depend on q (recall that we assumed q odd).

For a finite field $\mathbb{F}_q$, consider the finite symplectic group $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ with respect to the standard symplectic form

$$\begin{pmatrix} 0 & -I_N \\ I_N & 0 \end{pmatrix}.$$

We also now introduce the notation

$$(5) \quad \iota[B] := \begin{pmatrix} I_N & B \\ 0 & I_N \end{pmatrix}$$

defined by symmetric $N \times N$ matrices B . We call these elements the *advections* and call $\iota[B]$ a *simple advection* if B has rank 1. Two such elements $\iota[B]$ and $\iota[B']$ are conjugate in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ if and only if B and B' define equivalent symmetric bilinear forms on $\mathbb{F}_q^N$. The elements $\iota[B]$ in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ form the *Siegel unipotent* subgroup. The advections, and especially the simple advections, turn out to be the most difficult

cases of g for Theorem 1 and will be specially studied in all steps of the proof .

Our proof of Theorem 1 is organized as follows: First, we will prove that the contribution of the non-primitive representations in (4) has negative q -degree (see Theorem 7). The strategy is to split off a single type A (which we consider to include both the general linear groups and the non-split unitary groups) factor from the centralizer of the semisimple data of a non-primitive irreducible representation. The non-primitive representation can then be expressed as the Deligne-Lusztig induction of a representation of the Levi subgroup corresponding to the A -type factor, in which case we find a recursive formula for the character ratio of a Deligne-Lusztig induction. This is done in Section 3.

Since the number of primitive representations is constant, this reduces Theorem 1 to the claim that each individual character ratio for a primitive irreducible representation at a fixed non-central element is $o(1)$. In fact, this applies more generally and the individual character ratio of any non-trivial irreducible $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representation at a non-central element is $o(1)$ with respect to q (see Theorem 11). The proof of this is done in Section 4.

2. SET-UP

The purpose of this section is to recall the classification of irreducible representations for symplectic groups and state some set-up results. Specifically, the main purpose of this section is to recall the consistency of Lusztig classification data with Deligne-Lusztig induction from a Levi subgroup. We also note a special property of the primitive irreducible representations introduced above.

2.1. Notation for semisimple elements. We begin with some notation we will use in our discussion of the semisimple conjugacy class data. For a cyclic multiplicative group μ_n , let us fix an identification of μ_n with its Pontrjagin dual, writing ψ_a for the character associated to $a \in \mu_n$. In $\mathrm{GL}_r(\mathbb{F}_q) = \mathrm{U}_r^+(\mathbb{F}_q)$, for a choice of $\lambda \in \mathbb{F}_{q^r}^\times$ not in any lower field extension, we write A_λ^+ for the semisimple matrix with eigenvalues

$$(6) \quad \{\lambda, \lambda^q, \dots, \lambda^{q^{r-1}}\}$$

(in $\overline{\mathbb{F}_q}$). Its centralizer is

$$Z_{\mathrm{U}_r^+(\mathbb{F}_q)}(A_\lambda^+) = \mathrm{U}_1^+(\mathbb{F}_{q^r}) = \mathbb{F}_{q^r}^\times.$$

Similarly, in $U_r^-(\mathbb{F}_q)$, for a choice of $\lambda \in \mu_{q^{r+1}}$ (considered as the norm one elements of $\mathbb{F}_{q^{2r}}^\times$) not in $\mu_{q^{r'+1}}$ for any $r' < r$, we write A_λ^- for the semisimple matrix with eigenvalues (6) in $\overline{\mathbb{F}_q}$, with centralizer

$$Z_{U_r^-(\mathbb{F}_q)}(A_\lambda^-) = U_1^-(\mathbb{F}_{q^r}) = \mu_{q^{r+1}}.$$

We also write $A_\lambda^\pm$ for these elements' images under the embedding of $U_r^\pm(\mathbb{F}_q)$ in a symplectic or orthogonal group of rank r .

Now consider a semisimple conjugacy class $(s) \in \mathrm{SO}_{2N+1}(\mathbb{F}_q) = \mathrm{Sp}_{2N}^*(\mathbb{F}_q)$. We may choose the representative (s) of the form

$$s = I_{2p+1} \oplus (-I_{2\ell}) \oplus \bigoplus_i (A_\lambda^\pm)^{\oplus n_i}$$

for λ in $\mathbb{F}_{q^{r_i}}^\times$ or $\mu_{q^{r_i+1}}$ (and not for a lower field extension) such that the ranks add up, i.e. $N = p + \ell + \sum_i n_i r_i$. Its centralizer is then of the form

$$(7) \quad Z(s) = \mathrm{SO}_{2p+1}(\mathbb{F}_q) \times \mathrm{O}_{2\ell}^\pm(\mathbb{F}_q) \times \prod_i U_{n_i}^\pm(\mathbb{F}_{q^{r_i}}),$$

with identity component

$$(8) \quad Z(s)^\circ = \mathrm{SO}_{2p+1}(\mathbb{F}_q) \times \mathrm{SO}_{2\ell}^\pm(\mathbb{F}_q) \times \prod_i U_{n_i}^\pm(\mathbb{F}_{q^{r_i}}).$$

2.2. Lusztig classification data for a symplectic group. The irreducible representations of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ are then indexed by the data of

- (1) A semisimple conjugacy class $(s) \in \mathrm{SO}_{2N+1}(\mathbb{F}_q)$.
- (2) A unipotent representation u of $Z(s)^\circ$ obtained as a tensor product of symbols of each factor of (7) (allowing the non-irreducible degenerate symbol of the D -type factor in (8), if it occurs).
- (3) If $Z(s)$ is disconnected (in which case s has -1 eigenvalues and $Z(s)/Z(s)^\circ = \mathbb{Z}/2$) and the corresponding D -type symbol factor of u is non-degenerate, then the induction of u decomposes into two pieces

$$\mathrm{Ind}_{Z(s)}^{Z(s)^\circ}(u) = u^+ \oplus u^-$$

In this case, we also must choose sign data determining a summand of this induction. Otherwise, no additional data is needed and $(s), u$ specify an irreducible representation of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$.

For such $(s), u$, and possible sign data $\pm$, we write $r^{\mathrm{Sp}_{2N}}[(s), u, \pm]$ for the corresponding irreducible $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representation.

2.3. Splitting off a type A factor of a centralizer. Consider an irreducible $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representation corresponding to the data of (s) , u , and possible sign data $\pm$. Suppose $(s) \in \mathrm{SO}_{2N+1}(\mathbb{F}_q)$ can be expressed as

$$s = s' \oplus (A_\lambda^\pm)^{\oplus n}$$

for a choice of $\lambda \in \mu_{q^r+1}$ (and not for any lower field extension) and a semisimple element $s' \in \mathrm{SO}_{2N'+1}(\mathbb{F}_q)$ in which $\lambda, \lambda^q, \dots, \lambda^{q^{r-1}}$ are not eigenvalues. In particular, $N - N' = rn$ and we may factor s 's centralizer in $\mathrm{SO}_{2N+1}(\mathbb{F}_q)$ as

$$(9) \quad \begin{aligned} & Z_{\mathrm{SO}_{2N'+1}(\mathbb{F}_q)}(s') \times \mathrm{U}_n^\pm(\mathbb{F}_{q^r}) = \\ & Z_{\mathrm{SO}_{2N'+1}(\mathbb{F}_q)}(s') \times Z_{\mathrm{U}_n^\pm(\mathbb{F}_{q^r})}((A_\lambda^\pm)^{\oplus n}). \end{aligned}$$

Let us correspondingly factor the unipotent data u as $u = u' \otimes u_{\mathrm{U}_n^\pm}$ where u' is a unipotent representation of the centralizer of s' and $u_{\mathrm{U}_n^\pm}$ is a unipotent irreducible representation of $\mathrm{U}_n^\pm(\mathbb{F}_{q^r})$. We may consider the $\mathrm{Sp}_{2N'}(\mathbb{F}_q)$ -representation

$$r^{\mathrm{Sp}_{2N'}}[(s'), u', \pm]$$

and the $\mathrm{U}_n^\pm(\mathbb{F}_{q^r})$ -representation

$$u_{\mathrm{U}_n^\pm} \otimes \psi_\lambda(\det).$$

It turns out that the original representation of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ can be recovered from these two by taking their tensor product and applying Deligne-Lusztig induction.

Specifically, s 's centralizer (9) is a subgroup of $L(\mathbb{F}_q) = \mathrm{SO}_{2N'+1}(\mathbb{F}_q) \times \mathrm{U}_n^\pm(\mathbb{F}_{q^r})$, and in fact, we see that

$$Z_{\mathrm{SO}_{2N+1}(\mathbb{F}_q)}(s) = Z_{L(\mathbb{F}_q)}(s).$$

The group $L(\mathbb{F}_q)$ can be considered the fixed points of the standard Levi subgroup $L = \mathrm{SO}_{2N'+1} \times (\mathrm{GL}_n)^r$ over $\overline{\mathbb{F}_q}$ under a twisted (when $r > 1$ of the sign $\pm$ is $-$) Frobenius. In this case, writing L^* for the dual Levi subgroup of Sp_{2N} , the Deligne-Lusztig induction $R_{\mathrm{Sp}_{2N}}^{L^*}$ from this Levi subgroup with respect to the corresponding twisted Frobenius to $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ constructs the representations

$$\begin{aligned} & r^{\mathrm{Sp}_{2N}}[(s), u, \pm] = \\ & \pm R_{\mathrm{Sp}_{2N}}^{L^*}(r^{\mathrm{Sp}_{2N'}}[(s'), u', \pm] \otimes (u_{\mathrm{U}_n^\pm} \otimes \psi_\lambda(\det))) \end{aligned}$$

(see Theorem 7.1 (iv) and (vii) of [2]), with the sign $\pm$ arranged so that the right hand side has positive dimension. We write

$$|R_{\mathrm{Sp}_{2N}}^{L^*}(r^{\mathrm{Sp}_{2N'}}[(s'), u', \pm] \otimes (u_{\mathrm{U}_n^\pm} \otimes \psi_\lambda(\det)))|$$

for this sign multiple.

Comment: While it is tempting to work with a Levi factor with Frobenius fixed points

$$\mathrm{SO}_{2N'+1}(\mathbb{F}_q) \times \mathrm{U}_{nr}^{\pm}(\mathbb{F}_q)$$

instead of the Levi factor in Sp_{2N} described above since it has a simpler twisted Frobenius, this cannot be used in general since for r even, the non-split unitary group over the field extension is not a subgroup of any type A group over $\mathbb{F}_q$ of rank nr

$$\mathrm{U}_n^{-}(\mathbb{F}_{q^r}) \not\subseteq \mathrm{U}_{nr}^{\pm}(\mathbb{F}_q),$$

(though we may pass through this simpler Levi factor in the split case, since we do always have

$$\mathrm{GL}_n(\mathbb{F}_{q^r}) \subseteq \mathrm{GL}_{nr}(\mathbb{F}_q).$$

2.4. A property of the primitive representations. We also see that the character values of a primitive irreducible representation are independent of the specific values of the eigenvalues other than ± 1 of an input element. To state this precisely,

Definition 5. Consider G a symplectic or (special) orthogonal group over $\mathbb{F}_q$. We say two conjugacy classes $(g), (g') \in G$ have the same type if, considering their Jordan decompositions into commuting semisimple and unipotent factors $g = g_s \cdot g_u$, $g' = g'_s \cdot g'_u$, the semisimple parts g_s, g'_s have the same multiplicities of ± 1 eigenvalues,

$$Z(g_s) \cong Z(g'_s),$$

and there exists a bijection φ from the eigenvalues of g to the eigenvalues of g' such that $\varphi(\pm 1) = \pm 1$ and the unipotent summands g_u^{λ} and $g'_u{}^{\varphi(\lambda)}$ corresponding to each eigenvalue coincide. This defines an equivalence relation on conjugacy classes, and we denote the equivalence classes by $[g]$.

In particular,

Lemma 6. For a unipotent irreducible representation ρ of a finite reductive group G , if conjugacy classes (g) and (g') have the same type, then

$$\chi_{\rho}(g) = \chi_{\rho}(g').$$

For a primitive irreducible G -representation ρ , if conjugacy class $(g), (g')$ have the same type, then

$$|\chi_{\rho}(g)| = |\chi_{\rho}(g')|.$$

3. RECURSION USING DELIGNE-LUSZTIG INDUCTION

The purpose of this section is to prove the cancellation of character ratios of non-primitive irreducible representations of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ at all non-central elements.

Theorem 7. *For a non-central element $g \neq \pm I \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$, the character ratio sum over non-primitive irreducible representations is $o(1)$*

$$\lim_{q \rightarrow \infty} \sum_{\rho \in \widehat{\mathrm{Sp}_{2N}(\mathbb{F}_q)} \setminus \mathcal{P}} \frac{\chi_\rho(g)}{\dim(\rho)} = 0.$$

To prove this, let us introduce the following notation, which we will use throughout this section:

Fix a symplectic group $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ and a non-central $g \neq \pm I \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$. For a choice of $N' < N$, fix also an irreducible representation ρ' (which may be trivial) of $\mathrm{Sp}_{2N'}(\mathbb{F}_q)$ and a choice of natural numbers such that $nr = N - N'$. Let us write (s') for the semisimple data in $\mathrm{SO}_{2N'+1}(\mathbb{F}_q)$ corresponding to ρ' and consider

$$\mu_{q^r \pm 1}^{\mathrm{gen}} = \{\lambda \neq \pm 1 \in \mu_{q^r \pm 1} \mid \lambda \text{ is not an eigenvalue of } s'\} / \sim$$

under the equivalence relation where $\lambda \sim \lambda'$ when $A_\lambda^\pm$ and $A_{\lambda'}^\pm$ are conjugate. We may consider the Levi factors $L^\pm \subseteq \mathrm{Sp}_{2N}$ with respect to choices of Frobenius maps giving fixed points

$$L^\pm(\mathbb{F}_q) = \mathrm{U}_n^\pm(\mathbb{F}_{q^r}) \times \mathrm{Sp}_{2N'}(\mathbb{F}_q).$$

For a unipotent irreducible representation u of type A and rank n , we consider $\mathrm{U}_n^\pm(\mathbb{F}_{q^r})$ -representations $u_\lambda := u \otimes \psi_\lambda(\det)$.

Proposition 8. *Given the above notation, the q -degree of the sum of the character ratios*

$$(10) \quad \sum_{\lambda \in \mu_{q^r-1}^{\mathrm{gen}}} \frac{\chi_{R_{\mathrm{Sp}_{2N}}^{L^+}(\rho \otimes u_\lambda)}(g)}{\dim(R_{\mathrm{Sp}_{2N}}^{L^+}(\rho \otimes u_\lambda))} + \sum_{\lambda \in \mu_{q^r+1}^{\mathrm{gen}}} \frac{\chi_{|R_{\mathrm{Sp}_{2N}}^{L^-}(\rho \otimes u_\lambda)|}(g)}{\dim(|R_{\mathrm{Sp}_{2N}}^{L^-}(\rho \otimes u_\lambda)|)}$$

is strictly less than $2(nr - N)$ or strictly less than the q -degree of one of the character ratios of ρ'

$$\deg_q\left(\frac{\chi_{\rho'}(g')}{\dim(\rho')}\right)$$

at some non-central element $g' \in \mathrm{Sp}_{2N'}(\mathbb{F}_q)$.

Given this proposition, we can conclude the vanishing of the sum of character ratios for non-primitive irreducible representations:

Proof of Theorem 7. As described in Subsection 2.3, every representation ρ whose semisimple data (s) has a centralizer with connected component of the form

$$(11) \quad \mathrm{U}_{n_1}^{\pm}(\mathbb{F}_{q^{r_1}}) \times \cdots \times \mathrm{U}_{n_k}^{\pm}(\mathbb{F}_{q^{r_k}}) \times \mathrm{SO}_{2m+1}(\mathbb{F}_q) \times \mathrm{SO}_{2\ell}^{\pm}(\mathbb{F}_q)$$

can be constructed from a primitive irreducible representation ρ' by a process of k steps, where at each step the previous result is tensored with $(u_i)_{\lambda}$ for u_i an irreducible unipotent representation of A -type and rank n_i and $\lambda \in \mu_{q^{r_i} \mp 1}$, and then Deligne-Lusztig induction is applied to obtain a representation of the symplectic group of rank $n_1 r_1 + \cdots + n_k r_k + m + \ell$. Therefore, by the recursive behavior of the character ratios for Deligne-Lusztig inductions claimed in Proposition 8, we find that the sum of the character ratios at g of all such ρ obtained in this way for a fixed choice of (11) and u_i (over all possible choices of signs in the superscripts of each factor of (11) and all choices of λ_i) vanishes as q goes to infinity. Since the number of choices of subgroups H (up to isomorphism), irreducible unipotent representations u_i , and primitive irreducible representations ρ' does not grow with q , the sum of all these contributions is still $o(1)$ with respect to q , giving the claim. $\square$

The purpose of the remainder of the section is to prove Proposition 8. It is organized as follows: We begin with the case of $N - N' = 1$, $n = 1$, and $r = 1$, where the corresponding Levi factors have $\mathbb{F}_q$ -points $\mathrm{U}_1^{\pm}(\mathbb{F}_q) \times \mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$. We focus on the case of (signed) advections $g = \pm \iota[B]$ (recalling the notation (5)), with a special focus on the simple advection where B has rank 1. In fact, it will turn out that $\pm \iota[B]$ for B of rank 1 are the only cases of g where the bound of Proposition 8 is strict and cancellation between the different Deligne-Lusztig inductions in (10) is needed, making it the key case to discuss in the entirety of the proof. The case of $N - N' = 1$ is concluded in Subsection 3.1.

Next, we similarly consider the generalization to $N - N' = nr$ for $n = 1$ and $r > 1$, where the Levi factors have $\mathbb{F}_q$ -points $\mathrm{U}_1^{\pm}(\mathbb{F}_{q^r}) \times \mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$. This is done in Subsection 3.2.

Finally, all that remains are cases of $N - N' = nr$ for $n > 1$ and $r > 1$. No cancellation is needed in these cases, even for the simple advections. This is discussed in Subsection 3.3, concluding the proof of Proposition 8.

3.1. Splitting off a rank 1 factor over $\mathbb{F}_q$. We may approach the case of L^+ using elementary methods since it is a case of Harish-Chandra induction. In the case of L^- , we proceed by comparing the Deligne-Lusztig variety with the case of the Drinfeld curve.

Proof of Proposition 8 for $n = r = 1$. We begin with the case of $N' = N - 1$ and the case of $\iota[B]$. The case of $-\iota[B]$ is entirely similar, with the result of (10) at the signed advection being equal to a sign multiple of (10) at the true advection $\iota[B]$ (depending on the input representation ρ'). In the case of $g = \iota[B]$, (10) is

$$(12) \quad \sum_{(\lambda) \in \mu_{q-1}^{\text{gen}}} \frac{\chi_{R_{\text{Sp}_{2N}}^{L^+}(\rho' \otimes \psi_\lambda)}(\iota[B])}{\dim(R_{\text{Sp}_{2N}}^{L^+}(\rho' \otimes \psi_\lambda))} + \sum_{(\lambda) \in \mu_{q+1}^{\text{gen}}} \frac{\chi_{|R_{\text{Sp}_{2N}}^{L^-}(\rho' \otimes \psi_\lambda)|}(\iota[B])}{\dim(|R_{\text{Sp}_{2N}}^{L^-}(\rho' \otimes \psi_\lambda)|)}.$$

Since every $\iota[B]$ is unipotent, we can see that the character value of each summand of (12) will not depend on ψ , meaning that we must directly estimate each individual character ratio

$$\frac{\chi_{|R_{\text{Sp}_{2N}}^{L^\pm}(\rho' \otimes \psi)|}(\iota[B])}{\dim(|R_{\text{Sp}_{2N}}^{L^\pm}(\rho' \otimes \psi)|)}$$

and multiply the results by the number of possible ψ in the split and non-split cases (which are both $q/2$ plus a constant not depending on q). Our method is to show that the character ratio of these Deligne Lusztig inductions can be computed according to certain recursive formulae applied to character ratios of the input representation (see (16) and (32)). This can be done elementarily in the case of the split Levi factor L^+ , but requires computations with Deligne-Lusztig varieties in the case of the non-split Levi factor L^- , so we argue the two cases separately.

We from now on specifically consider the advection $\iota[B]$ for B the diagonal matrix $I_k \oplus 0_{N-k}$:

$$(13) \quad \left(\begin{array}{ccccccc} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & 0 & & & \\ & & & & \ddots & & \\ & & & & & 0 & \end{array} \right) \left. \vphantom{\begin{pmatrix} 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}} \right\}^k \left. \vphantom{\begin{pmatrix} 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}} \right\}^{N-k}$$

We denote this $N \times N$ matrix by $I_{k \subseteq N} = I_k \oplus 0_{N-k}$. (We only fix this to simplify notation, and we will describe the situation for the general case of B when it differs.)

The case of $\mathbf{GL}_1(\mathbb{F}_q)$: The Deligne-Lusztig variety is discrete for the split Levi factor L^+ , and the induction is just the Harish-Chandra induction

$$(14) \quad R_{\mathrm{Sp}_{2N}}^{L^+}(\rho' \otimes \psi) = \mathrm{Ind}_{\mathrm{Sp}_{2N}(\mathbb{F}_q)}^{P_1}(\rho' \otimes \psi)$$

where P_1 is the maximal parabolic subgroup (over $\mathbb{F}_q$) with Levi factor $\mathrm{Sp}_{2(N-1)}(\mathbb{F}_q) \times \mathrm{GL}_1(\mathbb{F}_q)$. The character of an induction can be calculated as

$$(15) \quad \chi_{\mathrm{Ind}_G^H(\rho')}(g) = \sum_{\substack{x \in G/H, \\ x^{-1}gx \in H}} \chi_{\rho'}(x^{-1}gx).$$

To apply this to compute the character of (14), write Λ_1 for the Lagrangian subspace preserved by P_1 . The condition of $x \in \mathrm{Sp}_{2N}(\mathbb{F}_q)/P_1$ satisfying $x^{-1}\iota[B]x \in P_1$ is equivalent to asking that $\iota[B]$ preserves the space $x\Lambda_1$, i.e. that $x\Lambda_1$ is contained in the 1-eigenspace $E_1(\iota[B]) = \mathbb{F}_q^{2N-k}$ of $\iota[B]$. These x 's are then identified with the space of lines in $E_1(\iota[B])$, of which there are a total of

$$\frac{q^{2N-k} - 1}{q - 1}.$$

For x corresponding to a line $L \subseteq E_1(\iota[B])$, we must compute the character of $\rho' \otimes \psi$ on the (Levi factor component) of the conjugate $x^{-1}\iota[B]x$. The $\mathrm{Sp}_{2(N-1)}(\mathbb{F}_q) \times \mathrm{GL}_1(\mathbb{F}_q)$ component of $x^{-1}\iota[B]x$ is of the form

$$(\iota[B']) \times (1) \in \mathrm{Sp}_{2(N-1)}(\mathbb{F}_q) \times \mathrm{GL}_1(\mathbb{F}_q)$$

where B' is obtained by restricting B to an $(N-1)$ -dimensional Lagrangian subspace. In the case of $B = I_{k \subseteq N}$, the only possibilities for the $(N-1) \times (N-1)$ matrix B' are

$$I_{k-1 \subseteq N-1} \quad \text{and} \quad I_{k \subseteq N-1}$$

(removing a single 1 or a single 0 diagonal entry from (13)). The number of outer conjugates giving these B' forms is

$$\frac{q^k - 1}{q - 1} \quad \text{and} \quad \frac{q^{2N-k} - q^k}{q - 1} = q^k \frac{q^{2(N-k)} - 1}{q - 1},$$

respectively. (For general B , there may be different cases of B' , however, writing k for the rank of B , there will still be exactly $\frac{q^k - 1}{q - 1}$ outer conjugates $x^{-1}\iota[B]x$ of the form $\iota[B']$ where B' has lower rank $k-1$, while all $q^k \frac{q^{2(N-k)} - 1}{q - 1}$ others are $\iota[B']$ for B' of rank k .)

We therefore obtain that the character of (14) at $\iota[B]$ is

$$\chi_{R_{\mathrm{Sp}_{2N}}^{L+}(\rho' \otimes \psi)}(\iota[I_{k \subseteq N}]) = \frac{q^k - 1}{q - 1} \cdot \chi_{\rho'}(\iota[I_{k-1 \subseteq N-1}]) + q^k \frac{q^{2(N-k)} - 1}{q - 1} \cdot \chi_{\rho'}(\iota[I_{k \subseteq N-1}]),$$

and therefore, the character ratio obeys the recursion

$$(16) \quad \frac{\chi_{R_{\mathrm{Sp}_{2N}}^{L+}(\rho' \otimes \psi)}(\iota[I_{k \subseteq N}])}{\dim(R_{\mathrm{Sp}_{2N}}^{L+}(\rho' \otimes \psi))} = \frac{q^k - 1}{q^{2N} - 1} \cdot \frac{\chi_{\rho'}(\iota[I_{k-1 \subseteq N-1}])}{\dim(\rho')} + q^k \frac{q^{2(N-k)} - 1}{q^{2N} - 1} \cdot \frac{\chi_{\rho'}(\iota[I_{k \subseteq N-1}])}{\dim(\rho')}.$$

The case of $\mathrm{U}_1^-(\mathbb{F}_q)$: To compute the non-split contribution, we must study the Deligne-Lusztig variety for the Levi subgroup L^- with respect to the twisted Frobenius. In other words, we must find the inverse image of the unipotent radical of the Levi subgroup under the Lang map

$$\mathcal{L} : \mathrm{Sp}_{2N}(\overline{\mathbb{F}_q}) \rightarrow \mathrm{Sp}_{2N}(\overline{\mathbb{F}_q})$$

sending an element to the composition of itself with the inverse of its image under the chosen Frobenius. Fix a choice of $\lambda \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^2$ and let us consider $\mathrm{U}_1^-(\mathbb{F}_q) \cong \mu_{q+1}$ as the group of matrices

$$(17) \quad \begin{pmatrix} x & y \\ \lambda y & x \end{pmatrix} \text{ such that } x^2 - \lambda y^2 = 1$$

for $x, y \in \mathbb{F}_q$. We note that the eigenvalues of this matrix are $x \pm \sqrt{\lambda}y$ with eigenvectors $(1, \pm\sqrt{\lambda})^T$, respectively. We change coordinates to consider $\mathrm{U}_1^-(\mathbb{F}_q)$ as the fixed points of the matrices (17) over $\overline{\mathbb{F}_q}$ under the (untwisted) Frobenius. A unipotent radical of the parabolic with Levi factor of matrices (17) in SL_2 consists precisely of matrices in SL_2 preserving, say, $(1, \sqrt{\lambda})^T$. In other words, the unipotent radical is a 1-dimensional non-rational Heisenberg group we will write as $\mathbb{H}_1$. In the case of $N = 1$, we can see that the inverse image of the matrices in SL_2 preserving $(1, \sqrt{\lambda})^T$ under $\mathcal{L}$ consists of matrices

$$(18) \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ such that } \begin{aligned} a^q + \sqrt{\lambda}b^q &= a + \sqrt{\lambda}b \\ c^q + \sqrt{\lambda}d^q &= c + \sqrt{\lambda}d \end{aligned}$$

By putting $x = a - \sqrt{\lambda}b$, $y = c - \sqrt{\lambda}d$ (since $(\sqrt{\lambda})^q = -\sqrt{\lambda}$), this recovers the Drinfeld curve

$$x^q y - y^q x = -2\sqrt{\lambda}.$$

We denote this by D . This is the Deligne-Lusztig variety corresponding to the embedding of $U_1^-(\mathbb{F}_q)$ in $SL_2(\mathbb{F}_q)$. Computing the cohomology, we find

$$\dim(H_c^k(D)) = \begin{cases} q^2 & \text{for } k = 1 \\ 1 & \text{for } k = 0 \\ 0 & \text{else} \end{cases}$$

From a homological perspective, the reduced homology $\tilde{H}_*(\tilde{D})$ of the unreduced suspension $\tilde{D}$ of D is concentrated in degree 2 (since D is 1-dimensional) where it is q^2 -dimensional. In fact, as a representation of the unipotent radical of $SL_2(\mathbb{F}_q)$, $\tilde{H}_2(\tilde{D})$ is

$$(19) \quad H_c^1(\tilde{D}) = \tilde{H}_2(\tilde{D}) \cong (\widetilde{R_{\mathbb{F}_q}})^{\oplus(q+1)} \oplus 1,$$

where $\widetilde{R_{\mathbb{F}_q}}$ denotes the reduced regular representation of $\mathbb{F}_q$, considered as the rational unipotent radical of $SL_2(\mathbb{F}_q)$. Each of the $q + 1$ copies corresponds to a character on $\mu_{q+1} = U_1^-(\mathbb{F}_q)$.

For the general case of N , consider the subgroup consisting of matrices

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

in Sp_{2N} preserving

$$(20) \quad (1, \underbrace{0, \dots, 0}_{N-1}, \sqrt{\lambda}, \underbrace{0, \dots, 0}_{N-1})^T.$$

This is the semidirect product

$$Sp_{2(N-1)} \ltimes \mathbb{H}_N$$

of a non-rational $(2N - 1)$ -dimensional Heisenberg group and the lower symplectic group. We change coordinates so that the unipotent radical of L^- as a Levi subgroup of Sp_{2N} is this Heisenberg group $\mathbb{H}_N$. Write $Y = \mathcal{L}^{-1}(\mathbb{H}_N)$ for the Deligne-Lusztig variety corresponding to L^- .

Claim 9. *The Deligne-Lusztig variety $Y = \mathcal{L}^{-1}(\mathbb{H}_N)$ satisfies the fibration*

$$(21) \quad Sp_{2(N-1)}(\mathbb{F}_q) \rightarrow Y \rightarrow \mathcal{D}$$

where $\mathcal{D}$ denotes the $(2N - 1)$ -dimensional analogue of D

$$(22) \quad \mathcal{D} = Z\left(\sum_{i=1}^N x_i^q y_i - x_i y_i^q = -2\sqrt{\lambda}\right)$$

(considered in $\mathbb{A}_{\mathbb{F}_q}^{2N} = \text{Spec}(\overline{\mathbb{F}_q}[x_1, \dots, x_N, y_1, \dots, y_N])$).

Proof of Claim 9. For a matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ to be an element of $\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N$ we first consider the condition that fixing (20) requires for the first columns of A , B , C , and D . Writing $\vec{a}$, $\vec{b}$, $\vec{c}$, and $\vec{d}$ for these first columns, we obtain the relations (18) with a , b , c , and d replaced by their corresponding vectors. Taking x_i to be the i th coordinate of $\vec{a} - \sqrt{\lambda}\vec{b}$ and taking y_i to be the i th coordinate of $\vec{c} - \sqrt{\lambda}\vec{d}$, similarly as in the case of SL_2 , this gives the condition (22).

Now, to prove the fibration (21), we first note that for the non-rational Heisenberg group $\mathbb{H}_N$, its intersection with its image $\mathbb{H}_N^F$ under the Frobenius is trivial

$$(23) \quad \mathbb{H}_N \cap \mathbb{H}_N^F = \{e\}.$$

(This follows since it is equivalent to proving

$$(\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N) \cap (\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N^F) = \text{Sp}_{2(N-1)},$$

which holds since, while $\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N$ can be interpreted as the subgroup of $\text{Sp}_{2N}(\mathbb{F}_q)$ fixing $(1, 0 \dots 0, \sqrt{\lambda}, 0, \dots, 0)^T$, the group $\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N^F$ is the subgroup of $\text{Sp}_{2N}(\mathbb{F}_q)$ fixing

$$\begin{aligned} & (1, 0 \dots 0, -\sqrt{\lambda}, 0, \dots, 0)^T = \\ & ((1, 0 \dots 0, \sqrt{\lambda}, 0, \dots, 0)^F)^T, \end{aligned}$$

so their intersection fixes a full 2-dimensional symplectic subspace and is therefore isomorphic to $\text{Sp}_{2(N-1)}$.)

Given this, we can see immediately a warm-up case of the fiber:

$$(24) \quad Y \cap (\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N) = \text{Sp}_{2(N-1)}(\mathbb{F}_q).$$

Specifically, for $g \in Y \cap (\text{Sp}_{2(N-1)} \ltimes \mathbb{H}_N)$, write $\bar{g}$ for its projection to $\text{Sp}_{2(N-1)}$. Since $(g^F)^{-1} \circ g \in \mathbb{H}_N$, we must have $(\bar{g}^F)^{-1} \circ \bar{g} = 1 \in \text{Sp}_{2(N-1)}$. Therefore, $\bar{g} \in \text{Sp}_{2(N-1)}(\mathbb{F}_q)$. Now write $g = \bar{g}h$ for an element $h \in \mathbb{H}_N$. In this case, then

$$(h^F)^{-1} \circ h = (h^F)^{-1} \circ \bar{g}^{-1} \circ \bar{g} \circ h = (g^F)^{-1} \circ g \in \mathbb{H}_N,$$

giving $h^F \in \mathbb{H}_N$. This implies that h is trivial by (23), therefore proving (24).

Now we consider the case of a general g in Y , i.e. so that $(g^F)^{-1} \circ g$ fixes (20). Let us write

$$\vec{v}_\pm = (1, 0, \dots, 0, \pm\sqrt{\lambda}, 0, \dots, 0)^T.$$

Note that $(\vec{v}_\pm)^F = \vec{v}_\mp$. To prove the fibration (21), it suffices to prove that there exists a unique element $h \in \mathbb{H}_N$ such that

$$(25) \quad g \circ h(\text{Span}(\vec{v}_+, \vec{v}_-)) = (g \circ h(\text{Span}(\vec{v}_+, \vec{v}_-)))^F.$$

Writing S for the symplectic form defining Sp_{2N} , we first note that

$$\begin{aligned} S(g^F(\vec{v}_-), g(\vec{v}_+)) &= S(g^F(\vec{v}_-), g^F(\vec{v}_+)) = \\ &= S(\vec{v}_-, \vec{v}_+) = S(g(\vec{v}_-), g(\vec{v}_+)). \end{aligned}$$

In particular,

$$S((g^F - g)(\vec{v}_-), g(\vec{v}_+)) = 0,$$

and therefore

$$S(g^{-1}((g^F - g)(\vec{v}_-)), \vec{v}_+) = 0.$$

We may hence pick the $h \in \mathbb{H}$ so that

$$h(\vec{v}_-) = \vec{v}_- + g^{-1}((g^F - g)(\vec{v}_-)).$$

In particular, for this choice of h , we obtain

$$(26) \quad g^F(\vec{v}_-) = g \circ h(\vec{v}_-).$$

Further, since $h \in \mathbb{H}$ preserves v_+ , its Frobenius h^F preserves v_- . In particular, the right-hand side of (25) can be re-written as

$$\begin{aligned} \text{Span}(g^F \circ h^F(\vec{v}_-), g^F \circ h^F(\vec{v}_+)) &= \text{Span}(g^F(\vec{v}_-), g^F \circ h^F(\vec{v}_+)) = \\ &= \text{Span}(g \circ h(\vec{v}_-), g^F \circ h^F(\vec{v}_+)). \end{aligned}$$

Comparing with the left-hand side

$$\text{Span}(g \circ h(\vec{v}_+), g \circ h(\vec{v}_-)),$$

it remains to prove that $g^F \circ h^F(\vec{v}_+)$ is a linear combination of $g \circ h(\vec{v}_+)$ and $g \circ h(\vec{v}_-) = g^F \circ h^F(\vec{v}_-)$. First, since $(g^F)^{-1} \circ g$ is an element of $\mathbb{H}$, it must preserve the symplectic space $\text{Span}(\vec{v}_+, \vec{v}_-)$ and we may also write

$$g^F(\vec{v}_-) = g(\vec{v}_-) + t \cdot g(\vec{v}_+)$$

for a constant t . Applying the Frobenius again gives

$$g^{F^2}(\vec{v}_+) = g^F(\vec{v}_+) + t^F \cdot g^F(\vec{v}_-),$$

which by (26) can be re-expressed as

$$g^F \circ h^F(\vec{v}_+) = g \circ h(\vec{v}_+) + t^F \cdot g \circ h(\vec{v}_-),$$

giving the claim. □

Given the fibration (21), since $\pi_1(\mathcal{D})$ is trivial, we in fact can write

$$(27) \quad Y \cong \mathrm{Sp}_{2(N-1)}(\mathbb{F}_q) \times \mathcal{D}.$$

Now, $\mathcal{D}$ has the homology of a join of N copies of the original Drinfeld curve

$$\mathcal{D} = \underbrace{D * \cdots * D}_N.$$

This can be proved by a spectral sequence coming from the filtration of Y by the corresponding Deligne-Lusztig varieties for symplectic subspaces

$$\mathbb{F}_q^{2N} \supset \mathbb{F}_q^{2(N-1)} \supset \cdots \supset \mathbb{F}_q^2.$$

Therefore, we find that the reduced homology of the unreduced suspension $\tilde{\mathcal{D}}$ of $\mathcal{D}$ is concentrated in degree $2N$ where

$$(28) \quad \dim(\tilde{H}_{2N}(\tilde{\mathcal{D}})) = q^{2N}.$$

Note that $\mathcal{D}$ is $(2N-1)$ -dimensional. Therefore, cohomologically, we obtain

$$\dim(H_c^k(\mathcal{D})) = \begin{cases} q^{2N} & \text{for } k = 2N-1 \\ 1 & \text{for } k = 0 \\ 0 & \text{else} \end{cases}$$

To construct $|R_{\mathrm{Sp}_{2N}}^{L^-}(\rho' \otimes \psi)|$ we take the alternating sum so that the $(2N-1)$ -degree cohomology has a positive coefficient. We can see that this recovers the dimension

$$|R_{\mathrm{Sp}_{2N}}^{L^-}(\rho' \otimes \psi)| = \frac{q^{2N} - 1}{q + 1} \dim(\rho').$$

(dividing by $q+1$ to pick out the ψ -isotypic component and multiplying by $\dim(\rho')$ for the ρ' -isotypic part of the group algebra contributed by the $\mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$ factor of (27)). To study the trace of g , we use (28) to understand the action of the unipotent subgroup of $\mathrm{SL}_2(\mathbb{F}_q)$ on $H_c^{2N-1}(\mathcal{D})$ as a tensor product of N copies of (19). Therefore, the ρ' -isotypic part of the alternating sum of Y 's cohomology consists of $(q^{2N} - 1)/(q^2 - 1)$ copies of the tensor product $\tilde{R}(\mathbb{F}_q) \otimes \rho'$, where:

- (A) For one copy of $\tilde{R}(\mathbb{F}_q) \otimes \rho'$, the factor of $\iota[I_{1 \subseteq N}]$ in $\mathrm{SL}_2(\mathbb{F}_q)$ acts as an advection and the factor in $\mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$ is trivial. In this case, the trace of the action on the corresponding summand of the $\rho' \otimes \psi$ -isotypic part of the cohomology is

$$(29) \quad \frac{1}{q+1} (q+1) \chi_{\tilde{R}(\mathbb{F}_q)} \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right) \chi_{\rho'}(I_{N-1}) = -\dim(\rho').$$

- (B) For $q^{2(N-1)} - 1$ of the $\tilde{R}(\mathbb{F}_q) \otimes \rho'$ copies, the factor of $\iota[I_{1 \subseteq N}]$ in $\mathrm{SL}_2(\mathbb{F}_q)$ acts as an advection and the factor in $\mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$ is $\iota[I_{1 \subseteq N-1}]$. The total contribution of the trace of the action on the corresponding summands of the ρ' -isotypic part of the cohomology in this case is

$$(30) \quad \frac{1}{q+1} (q^{2(N-1)} - 1)(q+1) \chi_{\tilde{R}(\mathbb{F}_q)} \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right) \chi_{\rho'}(\iota[I_{1 \subseteq N-1}]) = \\ - (q^{2N-2} - 1) \chi_{\rho'}(\iota[I_{1 \subseteq N-1}]).$$

- (C) For the remaining $(q^{2(N-1)} - 1)/(q^2 - 1)$ copies of $\tilde{R}(\mathbb{F}_q) \otimes \rho'$, the factor of $\iota[I_{1 \subseteq N}]$ corresponding to $\mathrm{SL}_2(\mathbb{F}_q)$ is trivial and the factor corresponding to $\mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$ is $\iota[I_{1 \subseteq N-1}]$. The total contribution of the trace of the action on the corresponding summands of the ρ' -isotypic part of the cohomology in this case is

$$(31) \quad \frac{1}{q+1} \frac{q^{2N-2} - 1}{q^2 - 1} (q+1) \chi_{\tilde{R}(\mathbb{F}_q)} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \chi_{\rho'}(\iota[I_{1 \subseteq N-1}]) = \\ \frac{q^{2(N-1)} - 1}{q+1} \chi_{\rho'}(\iota[I_{1 \subseteq N-1}]).$$

(In each case the factor of $1/q+1$ in (29), (30), and (31) occurs from considering the ψ -isotypic component on the cohomology). Therefore, we obtain

$$\chi_{|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi)|}(\iota[I_{1 \subseteq N}]) = -\dim(\rho') - q \frac{q^{2(N-1)} - 1}{q+1} \cdot \chi_{\rho'}(\iota[I_{1 \subseteq N-1}]).$$

and therefore, the character ratio obeys the recursion

$$(32) \quad \frac{\chi_{|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi)|}(\iota[I_{k \subseteq N}])}{\dim(|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi)|)} = \\ - \frac{q^k - (-1)^k}{q^{2N} - 1} \cdot \frac{\chi_{\rho'}(\iota[I_{k-1 \subseteq N-1}])}{\dim(\rho')} - q^k \frac{q^{2(N-k)} - 1}{q^{2N} - 1} \cdot \frac{\chi_{\rho'}(\iota[I_{k \subseteq N-1}])}{\dim(\rho')}.$$

We note that the analogue for a higher rank advection $\iota[I_{k \subseteq N}]$ is

$$\chi_{|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi)|}(\iota[I_{k \subseteq N}]) = \\ - \frac{q^k - (-1)^k}{q+1} \cdot \chi_{\rho'}(\iota[I_{k-1 \subseteq N-1}]) - q^k \frac{q^{2(N-k)} - 1}{q+1} \cdot \chi_{\rho'}(\iota[I_{k \subseteq N-1}]),$$

giving the character ratio obeys the recursion

$$(33) \quad \frac{\chi_{|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi)|}(\iota[I_{k \subseteq N}])}{\dim(|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi)|)} =$$

$$-\frac{q^k - (-1)^k}{q^{2N} - 1} \cdot \frac{\chi_{\rho'}(\iota[I_{k-1 \subseteq N-1}])}{\dim(\rho')} - q^k \frac{q^{2(N-k)} - 1}{q^{2N} - 1} \cdot \frac{\chi_{\rho'}(\iota[I_{k \subseteq N-1}])}{\dim(\rho')}.$$

Returning to the case of $g = \iota[I_{1 \subseteq N}]$, we must take the sum of (16) and (32) over all choices of ψ on $\mathrm{GL}_1(\mathbb{F}_q)$ and $\mathrm{U}_1^-(\mathbb{F}_q)$. Note that the values of (16) and (32) do not depend on ψ since $\iota[B]$ is unipotent, so we may interpret this sum as $|\mu_{q-1}^{\mathrm{gen}}|$ times (16) summed with $|\mu_{q+1}^{\mathrm{gen}}|$ times (32). The character ratios

$$\frac{\chi_{\iota[I_{k-1 \subseteq N-1}]}(\rho')}{\dim(\rho')} \text{ and } \frac{\chi_{\iota[I_{k \subseteq N-1}]}(\rho')}{\dim(\rho')}$$

all have non-positive q -degree. We note that therefore cancellation is only needed for our purpose in the case of $k = 1$, since for $k > 1$ the coefficients of (16) and (32) will already have negative q -degree even when multiplied by $|\mu_{q-1}^{\mathrm{gen}}|$ and $|\mu_{q+1}^{\mathrm{gen}}|$. In the case of $k = 1$, we must consider $|\mu_{q-1}^{\mathrm{gen}}|$ times the second term of (16) for $k = 1$ and $|\mu_{q+1}^{\mathrm{gen}}|$ times the second term of (32). These terms have opposite sign coefficients, and therefore the top terms will cancel, again leaving only contributions of strictly negative q -degree.

For any group element g not equal to a simple advection or its sign, the coefficients will not be large enough to give any terms of q -degree ≥ 0 , so cancellation is not needed. For example, in the case of Harish-Chandra induction, the sum of the coefficients appearing in the recursion is the number of 1-dimensional subspaces in an eigenspace of g , which then already has q -degree strictly less than $q^{2(N-1)}$. We see this explicitly in the case of $g = \iota[I_{k \subseteq N}]$ for $k > 1$ in (16) and (33). $\square$

3.2. Splitting off a rank 1 factor over a field extension $\mathbb{F}_{q^r}$. Before continuing the proof of Proposition 8, we first note a generalization of the discussion done in the $\mathrm{GL}_1(\mathbb{F}_q)$ -case of Subsection 3.1 to a larger example of Harish-Chandra induction. For convenience, let us use the notation

$$\binom{2n}{m}_{q, \mathrm{symp}} := \frac{|\mathrm{Sp}_{2n}(\mathbb{F}_q)|_{q'}}{|\mathrm{GL}_m(\mathbb{F}_q) \times \mathrm{Sp}_{2(n-m)}(\mathbb{F}_q)|_{q'}},$$

giving the number of m -dimensional isotropic subspaces of a symplectic space of dimension $2n$.

Lemma 10. *Write P_r for the maximal (rational) parabolic in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ with Levi factor $\mathrm{GL}_r(\mathbb{F}_q) \times \mathrm{Sp}_{2(N-r)}(\mathbb{F}_q)$. For a representation ρ of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ and a representation π of $\mathrm{GL}_r(\mathbb{F}_q)$, the character of the Harish-Chandra induction of $\rho \otimes \pi$ (obtained by applying the induction to the inflation of $\rho \otimes \pi$ to a representation of P_r by letting the unipotent factor act trivially) at the simple advection $\iota[I_{1 \subseteq N}]$ is*

$$(34) \quad \chi_{\mathrm{Ind}_{\mathrm{Sp}_{2N}}^{P_r}(\rho \otimes \pi)}(\iota[I_{1 \subseteq N}]) = \dim(\pi) \left(\binom{2(N-1)}{r-1}_{q, \mathrm{symp}} \dim(\rho) + q^r \binom{2(N-1)}{r}_{q, \mathrm{symp}} \chi_\rho(\iota[I_{1 \subseteq N-r}]) \right).$$

Proof. Let us consider the parabolic $P_r \subseteq \mathrm{Sp}_{2N}(\mathbb{F}_q)$ as the subgroup of elements preserving an r -dimensional isotropic subspace Λ of $\mathbb{F}_q^{2N}$. Recalling the induction formula (15), we find that the character of the Harish-Chandra induction at an element g can be expressed as

$$(35) \quad \chi_{\mathrm{Ind}_{\mathrm{Sp}_{2N}}^{P_r}(\rho \otimes \pi)}(g) = \sum_{\substack{x \in \mathrm{Sp}_{2N}(\mathbb{F}_q)/P_r \\ x^{-1}gx \in P_r}} \chi_\rho((x^{-1}gx)_{\mathrm{Sp}_{2(N-r)}}) \cdot \chi_\pi((x^{-1}gx)_{\mathrm{GL}_r})$$

where, for an element $y \in P_r$, we write y_{GL_r} and $y_{\mathrm{Sp}_{2(N-r)}}$ for the $\mathrm{GL}_r(\mathbb{F}_q)$ -factor and $\mathrm{Sp}_{2(N-r)}(\mathbb{F}_q)$ -factor of y 's Levi component, respectively.

Now, we may identify $\mathrm{Sp}_{2N}(\mathbb{F}_q)/P_r$ with the set of all r -dimensional isotropic subspaces in (by identifying $x \in \mathrm{Sp}_{2N}(\mathbb{F}_q)/P_r$ with $x\Lambda$). The condition that $x^{-1}gx \in P_r$ in (35) is equivalent to demanding that g preserves the isotropic space $x\Lambda$.

For an isotropic space $x\Lambda$ to be preserved by $\iota[I_{1 \subseteq N}]$, it must be contained in the $(2N-1)$ -dimensional 1-eigenspace of $\iota[I_{1 \subseteq N}]$. Let us write $e_1, \dots, e_N, f_1, \dots, f_N$ for the standard symplectic basis (so that $S(e_i, f_i) = 1$ and the symplectic form gives 0 on all other pairs of basis vectors) and assume without loss of generality that the $(2N-1)$ -dimensional space is the span of $e_2, \dots, e_N, f_1, \dots, f_N$. There are two cases of r -dimensional isotropic subspaces in a $(2N-1)$ -dimensional subspace of a symplectic space:

- (a) In the case where $f_1 \in x\Lambda$, the number of choices is equal to the number of $r-1$ -dimensional subspaces of the span of

$\text{Span}(e_2, \dots, e_N, f_2, \dots, f_N)$. The number of such $x\Lambda$ is then precisely

$$\binom{2(N-1)}{r-1}_{q,\text{symp}}.$$

- (b) If f_1 is not in $x\Lambda$, then the image of $x\Lambda$ under taking a quotient by f_1 gives an arbitrary r -dimensional isotropic subspace of $\text{Span}(e_2, \dots, e_N, f_2, \dots, f_N)$. For each choice of this projection, there are q^r choices of lifts to subspaces of the $(2N-1)$ -dimensional space $\text{Span}(e_2, \dots, e_N, f_1, \dots, f_N)$ (we obtain one factor of q for the possible $\mathbb{F}_q$ -coefficient of f_1 in each of the r basis vectors in $x\Lambda$), giving a total of

$$q^r \binom{2(N-1)}{r}_{q,\text{symp}}$$

choices.

Adding these contributions up, there are a total of

$$(36) \quad \binom{2(N-1)}{r-1}_{q,\text{symp}} + q^r \binom{2(N-1)}{r}_{q,\text{symp}}$$

terms. For $x\Lambda$ in case (a), both Levi factor components of the conjugate are trivial

$$(x^{-1}\iota[I_{1\subseteq N}]x)_{\text{Sp}_{2(N-r)}} = I_{2(N-r)}, \quad (x^{-1}\iota[I_{1\subseteq N}]x)_{\text{GL}_r} = I_r,$$

contributing the first term of (34). In case (b), we have

$$(x^{-1}\iota[I_{1\subseteq N}]x)_{\text{Sp}_{2(N-r)}} = \iota[I_{1\subseteq N-r}], \quad (x^{-1}\iota[I_{1\subseteq N}]x)_{\text{GL}_r} = I_r,$$

contributing the second term of (34). □

Lemma 10 will be applied to consider the case of Harish-Chandra induction, both in the case of rank 1 over a field extension and in all cases of higher rank. We consider the case of rank 1 over a field extension $\mathbb{F}_{q^r}$ now:

Proof of Proposition 8 for $n = 1$ and $r > 1$. In this case, we consider Levi factors

$$L^\pm(\mathbb{F}_q) = \text{U}_1^\pm(\mathbb{F}_{q^r}) \times \text{Sp}_{2(N-r)}(\mathbb{F}_q).$$

Again, we focus on the case of the simple advection $\iota[I_{1\subseteq N}]$, which is the only input element g where the bound will be strict. First, let us study the case of L^+ . In this case, we may also consider the intermediate Levi

factor $M^+ = \mathrm{GL}_r \times \mathrm{Sp}_{2(N-r)}$ with respect to the standard Frobenius so that

$$M^+(\mathbb{F}_q) = \mathrm{GL}_r(\mathbb{F}_q) \times \mathrm{Sp}_{2(N-r)}(\mathbb{F}_q).$$

Let us write P_r for the maximal parabolic over $\mathbb{F}_q$ in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ with $M^+(\mathbb{F}_q)$ as the Levi factor. In this case, the Deligne-Lusztig induction from L^+ to Sp_{2N} can be considered as the composition of the Deligne-Lusztig induction from $\mathrm{GL}_1(\mathbb{F}_{q^r})$ to $\mathrm{GL}_r(\mathbb{F}_q)$ with the Harish-Chandra induction from the rational parabolic P_r :

$$R_{\mathrm{Sp}_{2N}}^{L^+}(\rho' \otimes \psi_\lambda) = \mathrm{Ind}_{\mathrm{Sp}_{2N}}^{P_r}(\rho' \otimes R_{\mathrm{GL}_r(\mathbb{F}_q)}^{\mathrm{GL}_1(\mathbb{F}_{q^r})}(\psi_\lambda)).$$

Let us write

$$\rho'' := \rho' \otimes R_{\mathrm{GL}_r(\mathbb{F}_q)}^{\mathrm{GL}_1(\mathbb{F}_{q^r})}(\psi_\lambda).$$

As discussed above, the character of the Harish-Chandra induction from a parabolic of the form P_r obeys the recursion (34). Applying the dimension formula

$$\dim(R_{\mathrm{GL}_r(\mathbb{F}_q)}^{\mathrm{GL}_1(\mathbb{F}_{q^r})}(\psi_\lambda)) = \frac{|\mathrm{GL}_1(\mathbb{F}_{q^r})|_{q'}}{|\mathrm{GL}_r(\mathbb{F}_q)|_{q'}},$$

we may process (34) to obtain

$$(37) \quad \begin{aligned} & \chi_{R_{\mathrm{Sp}_{2N}}^{L^+}(\rho' \otimes \psi_\lambda)}(\iota[I_{1 \subseteq N}]) = \\ & (q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^{2(N-k+1)} - 1) \dim(\rho') + \\ & q^k \frac{(q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^{2(N-k)} - 1)}{q^k - 1} \chi_{\rho'}(\iota[I_{1 \subseteq N-r}]) \end{aligned}$$

Therefore, we obtain the recursion on character ratios

$$(38) \quad \begin{aligned} & \frac{\chi_{R_{\mathrm{Sp}_{2N}}^{L^+}(\rho' \otimes \psi_\lambda)}(\iota[I_{1 \subseteq N}])}{\dim(R_{\mathrm{Sp}_{2N}}^{L^+}(\rho' \otimes \psi_\lambda))} = \\ & \frac{q^r - 1}{q^{2N} - 1} + q^r \frac{q^{2(N-r)} - 1}{q^{2N} - 1} \frac{\chi_{\rho'}(\iota[I_{1 \subseteq N-r}])}{\dim(\rho')}. \end{aligned}$$

On the other hand, we may also consider the Deligne-Lusztig induction of L^- . Our strategy will be to first consider the case of $r = N$. In this case the Mackey formula (Theorem 11.13 of [3]) can be applied (by considering $\mathrm{U}_1^-(\mathbb{F}_{q^N})$ as a maximal torus over $\overline{\mathbb{F}_q}$ in Sp_{2N}) to obtain that

$$\chi_{|R_{\mathrm{Sp}_{2N}}^{\mathrm{U}_1^-(\mathbb{F}_{q^N})}(\psi_\lambda)|}(\iota[I_{1 \subseteq N}]) = -(q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^2 - 1)$$

(since, on the rational level, the intersection of the Levi factor with fixed points $\mathrm{SL}_{2(N-1)}(\mathbb{F}_q) \times \mathrm{GL}_1(\mathbb{F}_q)$ and $\mathrm{U}_1^-(\mathbb{F}_{q^N})$ is 0, though there is a double coset with non-trivial intersection over $\overline{\mathbb{F}_q}$). The general case can be obtained using similar methods as how the case of $\mathrm{U}_1^-(\mathbb{F}_q) \times \mathrm{Sp}_{2(N-1)}(\mathbb{F}_q)$ was obtained from the Drinfeld curve in the case of $N - N' = 1$ discussed in Subsection 3.1. In general, the character recursion is

$$(39) \quad \begin{aligned} & \chi_{|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi_\lambda)|}(\iota[I_{1 \subseteq N}]) = \\ & -(q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^{2(N-r+1)} - 1) \dim(\rho') + \\ & -q^r \frac{(q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^{2(N-r)} - 1)}{q^r + 1} \chi_{\rho'}(\iota[I_{1 \subseteq N-r}]). \end{aligned}$$

The second term's coefficient is obtained as the sum of two kinds of contributions with a term

$$(40) \quad -(q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^{2(N-r)} - 1)$$

playing the analogous role as the coefficient appearing in (30) and a term

$$(41) \quad \frac{(q^{2(N-1)} - 1)(q^{2(N-2)} - 1) \dots (q^{2(N-r)} - 1)}{q^r + 1}$$

playing an analogous role as (31). Again, for the case of $N = r$, this follows since we may apply Theorem 11.13 of [3]. For $N > r$, for the homology of the Deligne-Lusztig variety, we may, again, use the spectral sequence coming from the filtration of the variety by the corresponding Deligne-Lusztig varieties obtained by replacing $\mathbb{F}_q^{2N}$ by symplectic subspaces

$$\mathbb{F}_q^{2N} \supset \mathbb{F}_q^{2(N-1)} \supset \dots \supset \mathbb{F}_q^{2r}.$$

Since we are only interested in the Euler characteristic, we may work with the E^2 -term, in which the first summand of (39) and (41) correspond to trivial action of $\iota[I_{1 \subseteq N}]$, and (40) corresponds to a sum of copies of the reduced regular representation. From (39) then, we obtain the character ratio recursion

$$(42) \quad \begin{aligned} & \frac{\chi_{|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi_\lambda)|}(\iota[I_{1 \subseteq N}])}{\dim(|R_{\mathrm{Sp}_{2N}}^{L-}(\rho' \otimes \psi_\lambda)|)} = \\ & -\frac{q^r + 1}{q^{2N} - 1} - q^r \frac{q^{2(N-r)} - 1}{q^{2N} - 1} \frac{\chi_{\rho'}(\iota[I_{1 \subseteq N-r}])}{\dim(\rho')}. \end{aligned}$$

The number of terms in the character ratio sum (10) corresponding to L^+ and L^- each have q -degree r and the same top term, therefore giving the claim by canceling the second (and main) terms appearing in (38) and (42).

As in the case of $N - N' = 1$ discussed in Subsection 3.1, the case of $-\iota[I_{1 \subseteq N}]$ is similar (and in fact, only a sign multiple away from the case of $\iota[I_{1 \subseteq N}]$). For general g , the coefficients in both the Harish-Chandra of the character ratio terms are of strictly lower degrees, meaning that no cancellation is needed and the claimed bound is not strict. $\square$

3.3. Splitting of a factor of higher rank. The only remaining case of Proposition 8 to consider is for $N - N' = nr$ with $n > 1$. In this case, no cancellation of terms in (10) is needed, so that, in fact, the sum converges to 0 absolutely in this case. We will again use Lemma 10 and its proof to see this, for example, in the case of Harish-Chandra induction.

Proof of Proposition 8 for $n > 1$ and $r \geq 1$. This is the case of representations of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ with semisimple data with a full $\mathrm{GL}_n(\mathbb{F}_{q^r})$ or $\mathrm{U}_n^-(\mathbb{F}_{q^r})$ factor obtained by adding a generic eigenvalue $\lambda \in \mu_{q^r \mp 1}$ (and all Frobenius powers of it and its inverse) with multiplicity n to the input representation's semisimple data (s'). The number of choices of these generic eigenvalues has q -degree r . We claim that the q -degree of each character ratio

$$(43) \quad \frac{\chi_{|R_{\mathrm{Sp}_{2N}}^{L^\pm}(\rho' \otimes u_\lambda)|}(g)}{\dim(|R_{\mathrm{Sp}_{2N}}^{L^\pm}(\rho' \otimes u_\lambda)|)}$$

appearing in the sum (10) is less than or equal to $-(N - N') = -nr$ plus the q -degree of a character ratio or is less than or equal to $(n+1)r - 2N < 2(nr - N)$ (the latter bound only occurs in the case of $\iota[I_{1 \subseteq N}]$ when one of the character ratios in the recursion is the character ratio of the identity). Therefore, in the case of $n > 1$, no cancellation is needed and the claim follows immediately.

For example, let us consider the case of Harish-Chandra induction: Write $P_{N-N'}$ for the maximal parabolic containing the Levi factor $\mathrm{Sp}_{2N'}(\mathbb{F}_q) \times \mathrm{GL}_{N-N'}(\mathbb{F}_q)$ in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$. To consider the character ratio

$$\frac{\chi_{\mathrm{Ind}_{\mathrm{Sp}_{2N'}(\mathbb{F}_q)}^{P_{N-N'}}(\rho' \otimes \pi)}(g)}{\dim(\mathrm{Ind}_{\mathrm{Sp}_{2N'}(\mathbb{F}_q)}^{P_{N-N'}}(\rho' \otimes \pi))},$$

we recall Lemma 10 and its proof. Write Λ for the $(N - N')$ -dimensional isotropic subspace that $P_{N-N'}$ preserves. For a general g now, we consider the number of $(N - N')$ -dimensional isotropic subspaces $x\Lambda$ for $x \in \mathrm{Sp}_{2N}(\mathbb{F}_q)/P_{N-N'}$ preserved by g . It suffices to show the number of such $x\Lambda$ has q -degree less than or equal to

$$-(N - N') + \deg_q\left(\frac{|\mathrm{Sp}_{2N}(\mathbb{F}_q)|_{q'}}{|\mathrm{GL}_{N-N'}(\mathbb{F}_q) \times \mathrm{Sp}_{2N'}(\mathbb{F}_q)|_{q'}}\right),$$

since each x contributes a summand $\chi_{\rho' \otimes \pi}(x^{-1}gx)$, which we divide by

$$\dim(\mathrm{Ind}_{\mathrm{Sp}_{2N'}(\mathbb{F}_q)}^{P_{N-N'}}(\rho' \otimes \pi)) = \dim(\rho' \otimes \pi) \frac{|\mathrm{Sp}_{2N}(\mathbb{F}_q)|_{q'}}{|\mathrm{GL}_{N-N'}(\mathbb{F}_q) \times \mathrm{Sp}_{2N'}(\mathbb{F}_q)|_{q'}},$$

giving the claim. For a general g , $x\Lambda$ can only be preserved if it is a sum of isotropic subspaces of the eigenspaces, meaning that the case attaining the maximal number of choices is, again, $g = \iota[I_{1 \subseteq N}]$. As described in the proof of Lemma 10 (recalling (36)), for the simple advection, this number is

$$\binom{2(N-1)}{N-N'-1}_{q,\mathrm{symp}} + q^{N-N'} \binom{2(N-1)}{N-N'}_{q,\mathrm{symp}}.$$

By comparing numerators and denominators, we can calculate that the q -degree of each of these terms satisfy:

$$\deg_q\left(\binom{2(N-1)}{N-N'-1}_{q,\mathrm{symp}}\right) = \deg_q\left(\binom{2N}{N-N'}\right) - (N + N')$$

$$\deg_q(q^{N-N'} \binom{2(N-1)}{N-N'}_{q,\mathrm{symp}}) = \deg_q\left(\binom{2N}{N-N'}\right) - (N - N'),$$

giving the claim. The case for the non-split Levi factor analogously gives character ratios (43) of q -degree less by $-(N - N') = -nr$. $\square$

4. VANISHING OF THE INDIVIDUAL CHARACTER RATIOS

We have proved the vanishing of the partial character ratio sum over all non-primitive irreducible representations. To prove Theorem 1, it therefore remains to prove

$$\lim_{q \rightarrow \infty} \sum_{\rho \in \mathcal{P}} \frac{\chi_\rho(g)}{\dim(\rho)} = 0.$$

Since the number of primitive irreducible representations does not depend on q , it suffices to prove that each individual character ratio $\chi_\rho(g)/\dim(\rho)$ is $o(1)$ for $\rho \in \mathcal{P}$ and $g \neq \pm I \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$. In fact, in this section, we use elementary methods to prove that the individual character ratios of all $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representations are $o(1)$:

Theorem 11. *Fix a non-central element $g \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$. For every irreducible non-trivial representation ρ , the character ratio of ρ at g is $o(1)$*

$$(44) \quad \lim_{q \rightarrow \infty} \frac{\chi_\rho(g)}{\dim(\rho)} = 0.$$

To approach this, recall the natural embedding of $\mathrm{GL}_N(\mathbb{F}_q)$ as a Levi subgroup of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$

$$(45) \quad \begin{aligned} \mathrm{GL}_N(\mathbb{F}_q) &\xrightarrow{\subseteq} \mathrm{Sp}_{2N}(\mathbb{F}_q) \\ A &\mapsto \begin{pmatrix} A & 0 \\ 0 & (A^T)^{-1} \end{pmatrix}. \end{aligned}$$

We write P_N for the maximal parabolic subgroup of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ with this Levi factor. Its unipotent radical is precisely the Siegel unipotent of advections $\iota[B]$. Writing $\mathrm{Sym}^2(\mathbb{F}_q^N)$ for the abelian group of symmetric $N \times N$ matrices over $\mathbb{F}_q$ (with respect to $+$), the parabolic P_N can then be considered as the semidirect product

$$(46) \quad P_N = \mathrm{GL}_N(\mathbb{F}_q) \ltimes \mathrm{Sym}^2(\mathbb{F}_q^N),$$

with $A \in \mathrm{GL}_N(\mathbb{F}_q)$ acting on $M \in \mathrm{Sym}^2(\mathbb{F}_q^N)$ by congruence $A^T M A$.

Our approach to the proof of Theorem 11 is to first consider the special cases of non-central Levi group elements $g \in \mathrm{GL}_N(\mathbb{F}_q)$ and $g = \iota[B]$ for $B \neq 0 \in \mathrm{Sym}^2(\mathbb{F}_q^N)$. For the case of $g \in \mathrm{GL}_N(\mathbb{F}_q)$, we proceed by first proving an analogous claim for $\mathrm{GL}_N(\mathbb{F}_q)$ -representations and then using induction to pass first to P_N -representations and then to $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representations. We discuss this in Subsection 4.1. The case of $g = \iota[B]$ actually then follows immediately by an argument we discuss in Subsection 4.2, which serves as a warm-up case for the situation of general elements. Finally, we conclude the general case of Theorem 11 in Subsection 4.3.

4.1. The case of $\mathrm{GL}_N(\mathbb{F}_q)$ elements. The purpose of this subsection is to prove that Theorem 11 holds for non-central elements of $\mathrm{GL}_N(\mathbb{F}_q)$. Since we must approach this special case first, we restate it separately as the following

Proposition 12. *For a non-central element $g \in \mathrm{GL}_N(\mathbb{F}_q)$, considered as an element of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ by the natural embedding (45), for every irreducible non-trivial representation ρ the character ratio at g is $o(1)$ so that (44) holds.*

Our strategy is to begin with showing that the character ratios vanish for $\mathrm{GL}_N(\mathbb{F}_q)$ -representations not of the form $\psi(\det)$, since these character ratios can be estimated using elementary methods because the unipotent representations are all in the principal series:

Lemma 13. *For non-central $g \in \mathrm{GL}_N(\mathbb{F}_q)$ and any irreducible $\mathrm{GL}_N(\mathbb{F}_q)$ -representation ρ not of the form $\psi(\det)$ for a multiplicative character ψ of $\mathbb{F}_q^\times$, the corresponding character ratio $\chi_\rho(g)/\dim(\rho)$ is $o(1)$.*

Proof. Fix a non-central element $g \in \mathrm{GL}_N(\mathbb{F}_q)$. Consider the Lusztig classification data an irreducible representation ρ of $\mathrm{GL}_N(\mathbb{F}_q)$. Again, it corresponds to a semisimple conjugacy class (s) in the dual group $\mathrm{GL}_N^*(\mathbb{F}_q) = \mathrm{GL}_N(\mathbb{F}_q)$ and a unipotent irreducible representation u of the centralizer. We may assume by increasing q that all eigenvalues of the semisimple data are rational in $\mathbb{F}_q$. In this case, its centralizer is of the form

$$(47) \quad \mathrm{GL}_{n_1}(\mathbb{F}_q) \times \cdots \times \mathrm{GL}_{n_k}(\mathbb{F}_q),$$

where each factor corresponds to an eigenvalue λ_i of s of multiplicity n_i . In this case, factor the unipotent data u as a tensor product $u_1 \otimes \cdots \otimes u_k$ for unipotent irreducible $\mathrm{GL}_{n_i}(\mathbb{F}_q)$ -representations u_i . Then, writing P for the maximal parabolic of $\mathrm{GL}_N(\mathbb{F}_q)$ with Levi factor (47), ρ is the parabolic induction

$$\rho = \mathrm{Ind}_{\mathrm{GL}_N(\mathbb{F}_q)}^P \left(\bigotimes_{i=1}^k \psi_{\lambda_i}(\det) \otimes u_i \right)$$

(inflating the Levi factor representation by setting the unipotent radical to act trivially). The dimension of ρ is then $|\mathrm{GL}_N(\mathbb{F}_q)|/|P|$ times $\dim(u_1) \cdots \dim(u_k)$ and the character value of ρ at g is calculated using (47). Since every eigenspace of g has dimension strictly less than N , the character ratio is $o(1)$ in any case where (47) is not $\mathrm{GL}_N(\mathbb{F}_q)$ itself, i.e. when $\rho = \psi(\det) \otimes u$ for a multiplicative character $\psi \in \widehat{\mathbb{F}_q^\times}$ and an irreducible non-trivial unipotent representation u of $\mathrm{GL}_N(\mathbb{F}_q)$. Therefore, (for example, since, $|\psi(\det(g))| = 1$) it now suffices to prove the claim for $\rho = u$ a unipotent non-trivial irreducible representation.

We recall these are indexed by Young diagrams Λ with N boxes. We use the notation $\Lambda = (\lambda_1, \lambda_2, \dots)$ where λ_i is the length of the

i th row of Λ so that $\lambda_1 \geq \lambda_2 \geq \dots$. Write u_Λ for the corresponding unipotent irreducible representation. We recall that for two Young diagrams $\Lambda = (\lambda_1, \lambda_2, \dots)$ and $\Lambda' = (\lambda'_1, \lambda'_2, \dots)$, we say Λ is dominant over Λ' and write $\Lambda' \leq \Lambda$ if for all $k \geq 1$,

$$\lambda_1 + \lambda_2 + \dots + \lambda_k \geq \lambda'_1 + \lambda'_2 + \dots + \lambda'_k.$$

We use this to proceed by induction. Suppose that for every non-trivial Young diagram Λ strictly dominant over Λ the character ratio $\chi_{u_{\Lambda'}}(g)/\dim(u_{\Lambda'})$ is $o(1)$.

For a young diagram $\Lambda = (\lambda_1, \lambda_2, \dots, \lambda_n)$, let us consider the Levi subgroup

$$L_\Lambda = \mathrm{GL}_{\lambda_1}(\mathbb{F}_q) \times \mathrm{GL}_{\lambda_2}(\mathbb{F}_q) \times \dots \times \mathrm{GL}_{\lambda_n}(\mathbb{F}_q)$$

and a maximal parabolic $P_\Lambda \subseteq \mathrm{GL}_N(\mathbb{F}_q)$ containing it. Then the parabolic induction

$$(48) \quad \mathrm{Ind}_{\mathrm{GL}_N(\mathbb{F}_q)}^{P_\Lambda}(1)$$

decomposes as a sum of the irreducible unipotent u_Λ and copies of $u_{\Lambda'}$ for Λ' strictly dominant over Λ . (Note that the multiplicities of the dominant summands do not depend on q). Additionally, for a Young diagram Λ' strictly dominant over Λ ,

$$\deg_q(u_\Lambda) > \deg_q(u_{\Lambda'})$$

(this follows from the dimension formula for A -type symbols in [10]). Therefore, by the induction hypothesis, to prove the claim it suffices to show the character ratio for the whole induction (48) is $o(1)$

$$(49) \quad \lim_{q \rightarrow \infty} \frac{\chi_{\mathrm{Ind}_{\mathrm{GL}_N(\mathbb{F}_q)}^{P_\Lambda}(1)}(g)}{\dim(\mathrm{Ind}_{\mathrm{GL}_N(\mathbb{F}_q)}^{P_\Lambda}(1))} = 0.$$

The denominator is the Gaussian binomial coefficient $|\mathrm{GL}_N(\mathbb{F}_q)|/|P_\Lambda| = \binom{N}{\lambda_1 \dots \lambda_n}_q$, and the numerator is the number of outer conjugates of g in P_Λ ,

$$|\{x \in \mathrm{GL}_N(\mathbb{F}_q)/P_\Lambda \mid x^{-1}gx \in P_\Lambda\}|$$

(by the character formula (15)). This must have q -degree strictly less than $|\mathrm{GL}_N(\mathbb{F}_q)/P_\Lambda|$ since by the assumptions, all of g 's eigenspaces have dimension strictly less than N . Hence, (49) holds, which implies the claim for u_Λ , concluding the argument by induction. $\square$

We may easily pass to the analogous statement for character ratios of irreducible representations of P_N not of the form $\psi(\det)$ using the

induction formula, since the number of $\mathrm{GL}_N(\mathbb{F}_q)$ -orbits of symmetric matrices under congruence does not grow with q .

Lemma 14. *For non-central $g \in \mathrm{GL}_N(\mathbb{F}_q)$ and any irreducible representation ρ of P_N not of the form $\psi(\det)$ for a multiplicative character $\psi \in \widehat{\mathbb{F}_q^\times}$, the corresponding character ratio $\chi_\rho(g)/\dim(\rho)$ is $o(1)$.*

Proof. This follows recalling (46) because the number of conjugacy classes of P_N is a multiple of the number of conjugacy classes of the Levi factor not depending on q (since the number of symmetric bilinear forms over $\mathbb{F}_q$ does not depend on q). $\square$

We can now conclude the vanishing of character ratios of non-trivial $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representations at $\mathrm{GL}_N(\mathbb{F}_q)$ -representations, proving Proposition 12, i.e. Theorem 11 for $\mathrm{GL}_N(\mathbb{F}_q)$ elements not equal to $\pm I$.

Proof of Proposition 12. We first note that for each element $g \neq \pm I \in \mathrm{GL}_N(\mathbb{F}_q)$ which is central in $\mathrm{GL}_N(\mathbb{F}_q)$ is $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -conjugate to a non-central element of $\mathrm{GL}_N(\mathbb{F}_q)$ embedded in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$, and therefore we may fix without loss of generality a non-central element $g \in \mathrm{GL}_N(\mathbb{F}_q)$. Let us also fix a non-trivial irreducible representation ρ of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$. Consider the restriction of ρ to P_N . We consider two cases:

Case 1: The restriction $\mathrm{Res}_{P_N}(\rho)$ does not contain any representations of P_N of the form $\psi(\det)$. In this case, by Lemma 14, every irreducible summand $\pi \subseteq \mathrm{Res}_{P_N}(\rho)$ has $o(1)$ character ratio $\chi_\pi(g)/\dim(\pi)$ so that the q -degree of $\dim(\pi)$ is strictly greater than the q -degree of $\chi_\pi(g)$. In this case, the q -degree of $\dim(\rho)$ must then also be greater than the q -degree of $\chi_\rho(g)$, giving the claim.

Case 2: The restriction $\mathrm{Res}_{P_N}(\rho)$ contains a P_N -representation $\psi(\det)$. In this case, (since it is irreducible) ρ is a summand of the induction

$$\rho \subseteq \mathrm{Ind}_{\mathrm{Sp}_{2N}(\mathbb{F}_q)}^{P_N}(\psi(\det)).$$

Now we note that the multiplicity of representations of the form $\psi(\det)$ in

$$\mathrm{Res}_{P_N}(\mathrm{Ind}_{\mathrm{Sp}_{2N}(\mathbb{F}_q)}^{P_N}(\psi(\det)))$$

does not grow with P_N , by the classical Mackey formula. (This can also be seen since the dimension of

$$\begin{aligned} \mathrm{Hom}(1, \mathrm{Res}_{P_N}(\mathrm{Ind}_{\mathrm{Sp}_{2N}(\mathbb{F}_q)}^{P_N}(1))) &= \\ \mathrm{End}(\mathrm{Ind}_{\mathrm{Sp}_{2N}(\mathbb{F}_q)}^{P_N}(1)) \end{aligned}$$

does not depend on q since the decomposition of $\text{Ind}_{\text{Sp}_{2N}(\mathbb{F}_q)}^{P_N}(1)$ can be calculated according to the Littlewood-Richardson rule for symbols.) Therefore, the multiplicity of $\psi(\det)$ as a summand of $\text{Res}_{P_N}(\rho)$ does not grow with q . Since ρ is assumed to be non-trivial (and therefore $\dim(\rho)$ has positive q -degree), the non-trivial part of the restriction is non-zero. In other words, writing $\rho = \bigoplus_{\pi \in \widehat{P_N}} m_\pi \cdot \pi$, we have $m_\pi > 0$ for some $\pi \neq \psi(\det)$ while m_1 does not grow with q . We consider

$$(50) \quad \frac{\chi_\rho(g)}{\dim(\rho)} = \frac{\sum_{\pi \in \widehat{P_N}} m_\pi \chi_\pi(g)}{\sum_{\pi \in \widehat{P_N}} m_\pi \dim(\pi)}$$

noting that the main contribution of the denominator is for representations $\pi \neq \psi(\det)$ (which all have positive q -degree dimension). Therefore, (50) is $o(1)$ again by Lemma 14. $\square$

4.2. The case of advection elements. The statement of Theorem 11 for advection elements of the form $\iota[B]$ for $B \neq 0 \in \text{Sym}^2(\mathbb{F}_q^N)$ actually immediately follows from the case of $\text{GL}_N(\mathbb{F}_q)$ elements. We state and prove this case separately as a warm-up for the general case.

Proposition 15. *For a non-trivial element of the Siegel unipotent $\iota[B]$ corresponding to a non-zero $B \in \text{Sym}^2(\mathbb{F}_q^N)$, for every irreducible representation ρ , the character ratio at $\iota[B]$ is $o(1)$ so that (44) holds (for $g = \iota[B]$).*

Proof of Proposition 15. Suppose for an irreducible representation ρ and a non-trivial advection $\iota[B]$, the character ratio does not vanish, i.e. that (44) does not hold for $g = \iota[B]$. It suffices to prove that then ρ is trivial.

Since for congruent symmetric $N \times N$ matrices, the corresponding advectons are conjugate, we may assume without loss of generality that B is diagonal. Say B has non-zero diagonal entries $a_1, \dots, a_k$ for some $0 < k \leq N$ and that all others are 0, so that we may write

$$\iota[B] = \bigoplus_{i=1}^k \begin{pmatrix} 1 & a_i \\ 0 & 1 \end{pmatrix} \oplus I_{2(N-k)},$$

as an element of

$$(51) \quad \prod_{i=1}^k \text{SL}_2(\mathbb{F}_q) \times \text{Sp}_{2(N-k)}(\mathbb{F}_q),$$

embedded in $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ in the natural way by considering $\mathrm{SL}_2(\mathbb{F}_q)$ as the symplectic group on a two-dimensional space. In particular,

$$\chi_\rho(\iota[B]) = \chi_{\mathrm{Res}(\rho)}\left(\bigoplus_{i=1}^k \begin{pmatrix} 1 & a_i \\ 0 & 1 \end{pmatrix}\right)$$

where on the right-hand side, $\mathrm{Res}(\rho)$ denotes the restriction of ρ to the $\prod_{i=1}^k \mathrm{SL}_2(\mathbb{F}_q)$ factor of (51). This restriction decomposes as a sum of degree k tensor products of $\mathrm{SL}_2(\mathbb{F}_q)$ -representations. Let us write $m_{\kappa_1, \dots, \kappa_k}$ for the multiplicity of a tensor product $\kappa_1 \otimes \dots \otimes \kappa_k$ of $\mathrm{SL}_2(\mathbb{F}_q)$ -representations in the restriction, so that

$$\mathrm{Res}(\rho) = \bigoplus_{\kappa_i \in \widehat{\mathrm{SL}_2(\mathbb{F}_q)}} m_{\kappa_1, \dots, \kappa_k} (\kappa_1 \otimes \dots \otimes \kappa_k).$$

In particular, the character ratio for ρ can also be expressed as

$$(52) \quad \frac{\chi_\rho(\iota[B])}{\dim(\rho)} = \frac{\sum_{\kappa_i \in \widehat{\mathrm{SL}_2(\mathbb{F}_q)}} m_{\kappa_1, \dots, \kappa_k} \prod_{i=1}^k \chi_{\kappa_i} \left(\begin{pmatrix} 1 & a_i \\ 0 & 1 \end{pmatrix} \right)}{\sum_{\kappa_i \in \widehat{\mathrm{SL}_2(\mathbb{F}_q)}} m_{\kappa_1, \dots, \kappa_k} \prod_{i=1}^k \dim(\kappa_i)}.$$

As polynomials in q , it is known that for every non-trivial irreducible representation κ of $\mathrm{SL}_2(\mathbb{F}_q)$, the character value of any non-trivial element (in particular, any advection) on κ is strictly less than the q -degree of its dimension. Therefore, if (52) is not $o(1)$, then there must be a positive portion of true trivial representations $1 \otimes \dots \otimes 1$ in (52)

$$\lim_{q \rightarrow \infty} \frac{m_{1, \dots, 1}}{\sum_{\kappa_i \in \widehat{\mathrm{SL}_2(\mathbb{F}_q)}} m_{\kappa_1, \dots, \kappa_k}} > 0.$$

In this case, however, this the character ratio (52) will also fail to vanish if the advectons on the right hand side are replaced by any elements of $\mathrm{SL}_2(\mathbb{F}_q)$. For example, for a choice of distinct $\lambda_i \neq 1 \in \mathbb{F}_q^\times$, after replacing each advection $\begin{pmatrix} 1 & a_i \\ 0 & 1 \end{pmatrix}$ in (52) by $\begin{pmatrix} \lambda_i & 0 \\ 0 & \lambda_i^{-1} \end{pmatrix}$, the limit must still fail to vanish as q goes to infinity. In fact, considering the $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ element

$$g = \bigoplus_{i=1}^k \begin{pmatrix} \lambda_i & 0 \\ 0 & \lambda_i^{-1} \end{pmatrix} \oplus I_{2(N-k)}$$

(using the same natural embedding of (51)) the right hand side of (52) with the advections replaced by this torus element gives precisely ρ 's character ratio at g . Since g is the image of a non-central element of $\mathrm{GL}_N(\mathbb{F}_q)$ under the natural embedding (45), we can conclude that ρ must be trivial by Proposition 12. $\square$

4.3. Concluding the general case of Theorem 11. Now we conclude Theorem 11. First, as in the special cases of general linear group elements and the advections, we begin by proving the analogous claim for irreducible representations of the parabolic P_N .

Proposition 16. *Fix an irreducible representation ρ of P_N not of the form $\psi(\det)$ and fix a non-central element $g \in P_N$. The character ratio of ρ at g is then $o(1)$.*

Proof. Fix an irreducible P_N -representation ρ . Consider a general non-trivial element $g \in P_N = \mathrm{GL}_N(\mathbb{F}_q) \ltimes \mathrm{Sym}^2(\mathbb{F}_q^N)$. By Proposition 12 and Proposition 15, it only remains to consider cases of g whose factors in $\mathrm{GL}_N(\mathbb{F}_q)$ and $\mathrm{Sym}^2(\mathbb{F}_q^N)$ are both non-trivial.

First, let us fix an identification of the abelian group $\mathrm{Sym}^2(\mathbb{F}_q^N)$ with its Pontrjagin dual $\widehat{\mathrm{Sym}^2(\mathbb{F}_q^N)}$, writing ψ_M for the 1-dimensional irreducible representation corresponding to a symmetric matrix $M \in \mathrm{Sym}^2(\mathbb{F}_q^N)$. We consider the action of $\mathrm{GL}_N(\mathbb{F}_q)$ on $\widehat{\mathrm{Sym}^2(\mathbb{F}_q^N)}$ corresponding to its action by congruence on $\mathrm{Sym}^2(\mathbb{F}_q^N)$. We recall that the orbits of the action of $\mathrm{GL}_N(\mathbb{F}_q)$ on $\mathrm{Sym}^2(\mathbb{F}_q^N)$ precisely correspond to non-equivalent symmetric bilinear forms. Let us therefore write

$$\mathcal{O}_{r,\pm} = \{M \mid \mathrm{rank}(M) = r, \epsilon(\det(M)) = \pm\}$$

for the $\mathrm{GL}_N(\mathbb{F}_q)$ -orbit in $\mathrm{Sym}^2(\mathbb{F}_q^N)$ of matrices corresponding to symmetric bilinear forms of some rank $1 \leq r \leq N$ and discriminant $\pm$, so that

$$\mathrm{Sym}^2(\mathbb{F}_q^N) = \{0\} \amalg \coprod_{r=1}^N (\mathcal{O}_{r,+} \amalg \mathcal{O}_{r,-}).$$

Now let consider the restriction of ρ to $\mathrm{Sym}^2(\mathbb{F}_q^N)$. If this restriction only consists of trivial $\mathrm{Sym}^2(\mathbb{F}_q^N)$ -representations, then $\chi_\rho(g)$ is equal to the character value of ρ at the $\mathrm{GL}_N(\mathbb{F}_q)$ -component of g , in which case the character ratio must be $o(1)$ by Lemma 14.

Let us therefore assume that ρ restricts to a sum including at least one non-trivial $\text{Sym}^2(\mathbb{F}_q^N)$ -representation. We note that ρ must contain a full $\text{GL}_N(\mathbb{F}_q)$ -orbit's worth of characters ψ_M , and that for M and M' in the same orbit, the multiplicities of ψ_M and $\psi_{M'}$ in ρ match. Further, if ρ contains characters of $\text{Sym}^2(\mathbb{F}_q^N)$ corresponding to different $\text{GL}_N(\mathbb{F}_q)$ -orbits, we produce a proper subrepresentation by considering the subspace of ρ consisting only of a single orbit's worth of $\text{Sym}^2(\mathbb{F}_q^N)$ -characters. Therefore, by the irreducibility of ρ , it can only contain ψ_M for M in a single fixed $\text{GL}_N(\mathbb{F}_q)$ orbit. By the assumption of this case, this orbit is not $\{0\}$, since then ρ 's restriction to $\text{Sym}^2(\mathbb{F}_q^N)$ would be trivial. In summary, as a $\text{Sym}^2(\mathbb{F}_q^N)$ -representation, we may write

$$(53) \quad \bigoplus_{M \in \mathcal{O}_{r,\pm}} V_M \otimes \psi_M$$

for a choice of rank $1 \leq r \leq N$ and a choice of sign $\pm$ specifying an orbit $\mathcal{O}_{r,\pm}$ and writing V_M for the multiplicity vector space space of each ψ_M .

Now let us consider the set of all possible P_N -representations ρ' which restrict to the $\text{Sym}^2(\mathbb{F}_q^N)$ -representation (53). Such a ρ' is determined by a choice of how the action of $\text{GL}_N(\mathbb{F}_q)$ permutes the n copies of each character in the orbit. More specifically, to construct a representation ρ' , we must specify for each element $A \in \text{GL}_N(\mathbb{F}_q)$ and each $M \in \mathcal{O}_{r,\pm}$, an isomorphism

$$f_{A,M} \in \text{Hom}(V_M, V_{A^T M A})$$

such that the usual consistency diagrams

$$\begin{array}{ccc} V_M & \xrightarrow{f_{A,M}} & V_{A^T M A} \\ & \searrow f_{AB,M} & \downarrow f_{B,A^T M A} \\ & & V_{(AB)^T M (AB)} \end{array}$$

commute. Let us specifically consider the P_N -representation ρ_0 obtained by choosing for every choice of A and M ,

$$f_{A,M} = \text{Id}_N.$$

This representation may not be irreducible, however, we have $\dim(\rho_0) = \dim(\rho)$ and

$$(54) \quad |\chi_\rho(g)| \leq |\chi_{\rho_0}(g_{\text{GL}_N(\mathbb{F}_q)})|$$

for every choice of g since only the isotropy of $f_{g,M}$ will contribute to the trace of g on a $\text{GL}_N(\mathbb{F}_q)$ representation with restriction (53) and

by applying the triangle inequality to work with $g_{\mathrm{GL}_N(\mathbb{F}_q)}$. Therefore, it suffices to prove the vanishing of the character ratio

$$(55) \quad \lim_{q \rightarrow \infty} \frac{\chi_{\rho_0}(g_{\mathrm{GL}_N(\mathbb{F}_q)})}{\dim(\rho_0)} = 0.$$

To approach (55), we now consider the restriction of ρ_0 to $\mathrm{GL}_N(\mathbb{F}_q)$. Since the orbit $\mathcal{O}_{r,\pm}$ is not a singleton, the multiplicity of the trivial $\mathrm{GL}_N(\mathbb{F}_q)$ -representation in the restriction of ρ_0 does not grow with q , while the total dimension of ρ_0 does. As in the proof of Lemma 14, Lemma 13 can then be applied to conclude the vanishing of the character ratio. $\square$

The passage from P_N -representations to $\mathrm{Sp}_{2N}(\mathbb{F}_q)$ -representations is similar as in the proof of Proposition 12 above.

Proof of Theorem 11. Fix a non-central element $g \in \mathrm{Sp}_{2N}(\mathbb{F}_q)$. Again all symplectic group elements are conjugate to elements of P_N , so we may assume without loss of generality that g is an element of P_N . Fix an irreducible non-trivial representation ρ of $\mathrm{Sp}_{2N}(\mathbb{F}_q)$.

If the restriction of ρ to P_N does not contain any copies of $\psi(\det)$, then the claim follows immediately (as in Case 1 of the proof of Proposition 12) by Proposition 16.

If the restriction of ρ to P_N does contain some $\psi(\det)$, we consider ρ as a sub-representation of $\mathrm{Ind}_{\mathrm{Sp}_{2N}(\mathbb{F}_q)}^{P_N}(\psi(\det))$. As in Case 2 of the proof of Proposition 12, we then use the fact that the multiplicity of $\psi(\det)$ as a summand of the restriction of this induction does not grow with q to see that the multiplicity as a summand of $\mathrm{Res}_{P_N}(\rho)$ also does not grow with q . The remaining summands of $\mathrm{Res}_{P_N}(\rho)$ then overpower the contribution of $\psi(\det)$ in ρ 's character ratio at g , allowing us to conclude the vanishing again from Proposition 16. $\square$

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DEPARTMENT MATHEMATICS, PRINCETON UNIVERSITY, FINE HALL, 304 WASHINGTON RD, PRINCETON, NJ 08540